

# Nathan David Obeng-Amoako, E.I.T.

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## RESEARCH INTERESTS

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- Transit Signal Priority
- Transportation Safety Analysis
- Intelligent Transportation Systems
- Multimodal Corridor Planning
- Bicycle Network Analysis
- Geospatial Methods in Transportation

## EDUCATION

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**Doctor of Philosophy in Civil Engineering (Transportation)** Sep 2023 – Present

Northeastern University, Boston, MA (GPA: 4.00/4.00)

**Research Advisor:** Professor Peter Furth

**Master of Science in Civil Engineering (Transportation)** Sep 2023 – Apr 2025

Northeastern University, Boston, MA (GPA: 4.00/4.00)

**Bachelor of Science in Civil Engineering** Aug 2018 – Sep 2022

Kwame Nkrumah University of Science and Technology (KNUST), Kumasi

**Recognition:** Best Graduating Student, Class of 2022

## PROFESSIONAL EXPERIENCE

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**Transportation Engineering Intern, Fehr & Peers, Oakland, CA** Apr 2026 – Present

- Supporting multimodal transportation engineering and planning projects including transit signal priority, complete streets analysis, nighttime safety planning, and BART station area studies across the Bay Area.
- Applying GIS analysis and active transportation planning methods across corridor, safety, and transit-oriented development contexts for municipal and regional clients, including grant application inputs.

**Assistant Transportation Engineer, BHM Construction International UK Ltd.** Oct 2022 – July 2023

- Conducted over 20 quality control tests daily on construction materials to ensure strict standard compliance
- Supported site reporting and coordination with a team of 5 engineers to improve workflow efficiency

**Civil Engineering Trainee, China Henan International Corporation Ltd.** Sep 2021 – Dec 2021

- Supervised 15+ artisans daily to ensure strict adherence to project specifications and timelines
- Contributed to precise structural detailing for reinforced concrete components

## SPONSORED RESEARCH PROJECTS

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**Project:** Bicycle Level of Traffic Stress at Intersections (xLTS) Jun 2024 – Aug 2025

**Funding Agency:** United States Department of Transportation (USDOT)

- Assessing the impact of high-stress crossings on bike network connectivity
- Developing algorithms and data structures for traffic stress network analysis

## SPONSORED RESEARCH PROJECTS (CONT'D)

**Project:** Accessible Bus Stop Design in the Presence of Bike Lanes

**Sep 2023 – Jun 2024**

**Funding Agency:** Massachusetts Department of Transportation (MassDOT)

- Studied bus rider and bicyclist behavior at floating and constrained bus stops
- Proposed design improvements to reduce bus-rider-and-bicyclist conflicts

## PEER-REVIEWED JOURNAL PUBLICATIONS

- 2025
1. Peter Furth, Ray Saeidi-Razavi, **Nathan David Obeng-Amoako** and Milad Tahmasebi. TSP-Friendly Underlying Traffic Signal Control: An Essential Complement of Transit Signal Priority. *Future Transportation* (2025), doi: <https://doi.org/10.3390/futuretransp5040160>
  2. Joshua Kofi Asamoah, Blessing Agyei Kyem, **Nathan David Obeng-Amoako**, Armstrong Aboah, SAAM-ReflectNet: Sign-Aware Attention-Based Multitasking Framework for Integrated Traffic Sign Detection and Retroreflectivity Estimation, *Expert Systems With Applications* (2025), doi: <https://doi.org/10.1016/j.eswa.2025.128003>

### Submitted and under review:

- 2026
3. Yu-Min (Thomas) Yanga, Dewan Tanvir Ahammed, **Nathan David Obeng-Amoako**, Chengbo Ai, Peter Furth and Eleni Christofa. Community-Engaged Informed Floating Bus Stop Guidelines. *Transportation Research Record*.

## CONFERENCE PRESENTATIONS

- 2026
1. Yu-Min (Thomas) Yanga, Dewan Tanvir Ahammed, **Nathan David Obeng-Amoako**, Eleni Christofa, Chengbo Ai, and Peter Furth Community-Engaged Informed Floating Bus Stop Guidelines. *Transportation Research Board Annual Meeting 2026*, Washington, D.C.
- 2025
2. **Nathan David Obeng-Amoako** and Peter Furth. (May 2024). Underlying Traffic Signal Control: The Key to Effective Transit Signal Priority – A Case Study of MBTA Bus Route 39 along South Huntington Avenue, Boston, MA. *2025 MassDOT Transportation Innovation Conference*, Worcester, MA.
  3. Yu-Min (Thomas) Yanga, Dewan Tanvir Ahammed, **Nathan David Obeng-Amoako**, Eleni Christofa, Chengbo Ai, and Peter Furth. Design Guidelines for Accessible Floating Bus Stops. *International Conference on Transportation and Development 2025*, ASCE, Glendale, AZ.
- 2024
4. Yu-Min (Thomas) Yanga, Dewan Tanvir Ahammed, **Nathan David Obeng-Amoako**, Eleni Christofa, Chengbo Ai, and Peter Furth. (April 2024). Accessible Bus Stop Design in the Presence of Bike Lanes. Poster. *2024 MassDOT Transportation Innovation Conference*, Worcester, MA.
- 2023
5. **Nathan David Obeng-Amoako**, Arthur Louis Senaya, Bundu Kassim, Francisca Owusu-Ansah, Eugene Damoah, Santus Worlachie, and Kenneth A. Tutu. (April 2023). Modification of Bitumen with Waste Materials for Enhanced Aggregate Retention in Surface-Dressed Roads. *Lectern. 3rd IRF Africa Regional Congress & Exhibition*, Accra, Ghana.

## RESEARCH REPORTS

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|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 | 1. Eleni Christofa, Chengbo Ai, Peter Furth, Yu-Min Yang, Dewan Tanvir Ahammed, and <b>Nathan David Obeng-Amoako</b> . (2024). <i>Accessible Bus Stop Design in the Presence of Bike Lanes</i> . Massachusetts Department of Transportation (MassDOT), Office of Transportation Planning. Report No. 24-060. Retrieved from <a href="https://rosap.ntl.bts.gov/view/dot/79327">https://rosap.ntl.bts.gov/view/dot/79327</a> |
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## JOURNAL REVIEWS

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|---------------------------------------------------------------------|-----------------------|
| 1. Journal of Transportation Engineering, Part A: Systems, ASCE     | <b>2026 – Present</b> |
| 2. Transportation Research Interdisciplinary Perspectives, Elsevier | <b>2025 – Present</b> |
| 3. Expert Systems with Applications, Elsevier                       | <b>2025 – Present</b> |
| 4. Transportation Research Record, SAGE Publications                | <b>2024 – Present</b> |

## NOTABLE DISTINCTIONS AND ACHIEVEMENTS

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|----------------------------------------------------------------------------------------|--------------------|
| • Grant Recipient, New England African Business Expo, Worcester, MA                    | <b>May 2025</b>    |
| • Civil & Environmental Engineering Distinguished Fellowship Award, Northeastern Univ. | <b>Aug 2023</b>    |
| • Associated Consultants Ltd. Award for Best Graduating Student, KNUST                 | <b>Jul 2023</b>    |
| • Provost Award for Best & Excellent Students in College of Engineering, KNUST (x4)    | <b>2019 – 2022</b> |
| • Academic Excellence Award, Ghana Engineering Students' Association, Ghana            | <b>Sep 2022</b>    |
| • Global Finalist, Unilever Future Leaders League Business Pitch Competition, U.K.     | <b>Jul 2022</b>    |
| • Scholar, Tullow Oil Scholarship Scheme, Ghana                                        | <b>Apr 2019</b>    |
| • Semifinalist, National Science and Maths Quiz (NSMQ), Ghana                          | <b>Jul 2018</b>    |

## RELEVANT SOFTWARE AND SKILLS

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|------------------------------|----------------------------------------------------------------------------|
| • <b>Python, R, MATLAB</b>   | machine learning, deep learning, advanced data analysis                    |
| • <b>PTV Vissim/Vistro</b>   | traffic flow simulations, traffic signal analysis, traffic impact analysis |
| • <b>Synchro</b>             | traffic flow modeling, traffic signal timing optimization                  |
| • <b>ArcGIS</b>              | spatial data mapping and analysis for informed decision-making             |
| • <b>Gephi</b>               | network data analysis and visualization                                    |
| • <b>AutoCAD</b>             | precision drafting and technical drawings, 2D and 3D                       |
| • <b>HTML/CSS/JavaScript</b> | developing web applications                                                |
| • <b>Minitab, MS Excel</b>   | data visualization, statistical computing, data organization               |

## APPOINTMENTS AND ADVISORY ROLES

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|-----------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <b>Member, Massachusetts Bicycle and Pedestrian Advisory Board, MassDOT</b>                                                 | <b>Jan 2026 – Present</b> |
| • Appointed by Governor Maura T. Healey to advise on state-wide bicycle and pedestrian infrastructure policies and programs |                           |
| • Provide guidance on transportation infrastructure, safety, and accessibility initiatives                                  |                           |

## LEADERSHIP EXPERIENCE

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| <b>Secretary, Ghana Transportation Professionals Forum (GTPF)</b>                                                                                                                                                              | <b>Feb 2026 – Present</b>  |
| <ul style="list-style-type: none"><li>• Maintain membership records and coordinate executive board communications</li><li>• Support event logistics and engagements with Ghana transportation stakeholders</li></ul>           |                            |
| <b>Committee Member, Career Mixer Committee, CEE Dep't, Northeastern Univ.</b>                                                                                                                                                 | <b>Sep 2025 – Feb 2026</b> |
| <ul style="list-style-type: none"><li>• Organizing networking events connecting students with industry</li><li>• Coordinating logistics for career dev't workshops and employer engagements</li></ul>                          |                            |
| <b>Executive C'ttee Member, CEE Graduate Student Council, Northeastern Univ.</b>                                                                                                                                               | <b>Feb 2024 – Present</b>  |
| <ul style="list-style-type: none"><li>• Brainstorming and implementing strategies to enhance graduate student experience</li><li>• Serving as a liaison between the departmental chair and the graduate student body</li></ul> |                            |
| <b>Department Rep., College of Engineering. PhD Council, Northeastern Univ.</b>                                                                                                                                                | <b>Jun 2024 – Aug 2025</b> |
| <ul style="list-style-type: none"><li>• Meeting with advisor to discuss novel ways of improving student experience</li><li>• Organizing weekly coffee hours and semesterly happy hours for PhD students</li></ul>              |                            |
| <b>Communications Director, Young Professionals and Youth Coalition, Ghana</b>                                                                                                                                                 | <b>Jan 2021 – Dec 2021</b> |
| <ul style="list-style-type: none"><li>• Social media management to curate consistent communications content</li><li>• Organization of in-person and virtual engagements with members and partners</li></ul>                    |                            |
| <b>President, Civil Engineering Students' Association, KNUST</b>                                                                                                                                                               | <b>Feb 2020 – Sep 2021</b> |
| <ul style="list-style-type: none"><li>• Coordination of fundraising events to secure administrative equipment</li><li>• Establishment of academic support scheme for underachieving students</li></ul>                         |                            |

## TEACHING EXPERIENCE

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| <b>Northeastern University, Boston, MA</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Summers of 2024, 2025, 2026                | <b>Program Assistant, Sustainable Transportation Summer School Program</b><br>Location: Technische Universiteit Delft, The Netherlands <ul style="list-style-type: none"><li>• Facilitate logistical, academic and evaluative support for an intensive 5-week program</li><li>• Assist in teaching Vision Zero safety principles to 25 students</li><li>• Explore strategies for car-free and car-lite zones to enhance active transportation</li></ul> |
| Spring 2026                                | <b>Graduate Teaching Assistant, Energy Systems: Science, Technology, and Sustainability</b> <ul style="list-style-type: none"><li>• Provided grading support for coursework on energy technologies and sustainability</li><li>• Gave an invited lecture on Transportation Energy Systems in a Developing World</li></ul>                                                                                                                                |
| Spring 2026                                | <b>Graduate Teaching Assistant, Transportation Systems Analysis and Planning</b> <ul style="list-style-type: none"><li>• Supported students during office hours with questions on design software (Synchro)</li><li>• Evaluated assignments on intersection analysis, signal timing, and street design principles</li></ul>                                                                                                                             |
| Fall 2025                                  | <b>Graduate Teaching Assistant, Highway Design</b> <ul style="list-style-type: none"><li>• Provided grading support for homework assignments</li><li>• Assisted students during office hours, addressing questions about design principles</li></ul>                                                                                                                                                                                                    |

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| Fall 2025 | <b>Graduate Teaching Assistant, Transportation Systems Analysis and Planning</b> <ul style="list-style-type: none"> <li>• Evaluated coursework and provided constructive feedback on analytical assignments</li> <li>• Supported students' understanding of transportation system planning frameworks</li> </ul> |
| Fall 2023 | <b>Course Instructor, Recitation for Statics and Solid Mechanics</b> <ul style="list-style-type: none"> <li>• Tutored fundamental concepts of Newtonian physics and solid mechanics</li> <li>• Evaluated students' test performance and offered comprehensive feedback</li> </ul>                                |

## PROFESSIONAL AFFILIATIONS

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|-------------------------------------------------------------|---------------------------|
| • American Society of Civil Engineers (ASCE)                | <b>Feb 2025 – Present</b> |
| • Boston Society of Civil Engineers Section (BSCES) of ASCE | <b>Feb 2025 – Present</b> |
| • New England Intelligent Transportation Society (NEITS)    | <b>Sep 2023 – Present</b> |
| • Institute of Transportation Engineers (ITE)               | <b>Jan 2024 – Present</b> |

## VOLUNTEER EXPERIENCE & COMMUNITY INVOLVEMENT

- **Judge**, Massachusetts Science and Engineering Fair, Worcester (April 2025)
- **Cycling Activist**, Boston Cycling Union, Boston, MA (Feb 2025 – Present)
- **Executive Vice President**, Student Energy Society, KNUST (Jan 2022 – Oct 2022)
- **Community Outreach Coordinator**, Nat'l Society of Black Engineers, KNUST (Jan 2022 – Sep 2022)
- **Project Coordinator**, Engineers Without Borders, KNUST (Aug 2021 - Jul 2022)

## BLOG POSTS

1. "Ghana is Moving, but Barely: The Case for a Smarter Public Transportation Strategy." MyJoyOnline, April 2026. <https://www.myjoyonline.com/ghana-is-moving-but-barely-the-case-for-a-smarter-public-transportation-strategy/>
2. "Replacing the Reid Overpass with an At-Grade Intersection (Cambridge, near BU Bridge)." Co-authored with Peter G. Furth. January 2026. <https://peterfurth.sites.northeastern.edu/2026/01/06/replacing-the-reid-overpass-with-an-at-grade-intersection-cambridge-near-bu-bridge/>
3. "PhD Student Reflects on the Sustainable Transportation Dialogue of Civilizations in Holland." Northeastern University College of Engineering News, August 2024. <https://coe.northeastern.edu/news/a-reflection-on-the-sustainable-transportation-dialogue-of-civilizations-course-in-holland/>
4. "Bus-Friendly Traffic Signals Can Reduce Bus Delay By 90% on South Huntington Avenue." Co-authored with Peter G. Furth. April 2024. <https://peterfurth.sites.northeastern.edu/2024/06/04/bus-friendly-traffic-signals-can-reduce-bus-delay-by-90-on-south-huntington-avenue/>

## REFERENCES

*Available upon request*